

4 THE VIRTUAL UNIT LOAD METHOD

4.1 INTRODUCTION

4.1.1 Scope

The methods covered so far (Direct Integration, Macaulay's, Conjugate beam, Moment-Area) are only applicable to **normal beams** subject to **normal loading**. If the structure has members which lie in more than one longitudinal axis, such as in frames, these methods become severely handicapped. There is need, therefore, to explore a more powerful method which can handle both beams and frames.

The **Virtual Unit Load** method is applicable not only to **beams** and **rigid jointed frames**, but also to **trusses**, **hybrid frames** and even **arches**. The method is therefore one of the most powerful and versatile tools in structural analysis.

Virtual means imaginary, or fictitious, or **unreal**. The term **displacement** means a change in position, or a change in orientation. The term **deflection** is used to refer to the change in position, while the term **slope (or rotation)** is used to refer to the change in orientation.

4.1.2 Basis

The Virtual Unit Load method is based on the **principle of conservation of energy** which was developed by **Daniel Bernoulli © 1738**, based on the **principle of virtual work**, which was developed by **Johann Bernoulli © 1715**. (Johann was the father of Daniel). Virtual work is defined as the work done by a **real force system** acting on a body, which is in static equilibrium, when the system is given some **virtual displacement**. [Virtual work may also be defined as the work done by a **virtual force system** acting on a body, which is in static equilibrium, when the system is given some **real displacement**].

The principle of conservation of energy then dictates that “**For a system of forces acting on a body, the body is in static equilibrium if, when it is given any virtual displacement, the net work done by the system of forces is zero**”.

Thus, External work done = Internal work done.

4.1.3 Internal Work

In **flexural structures** (i.e. beams and rigid-jointed frames), the internal work done is stored as strain energy due to flexural, shear, and axial strains. However, the strain energy due to shear and axial strains is negligible compared to the strain energy due to flexural strains. Therefore, only **flexural strains** will be considered in determining displacements in flexural structures.

In **trusses**, the members are subjected to axial forces only, and the internal work done is stored as strain energy due to axial strains. Therefore, only **axial strains** will be considered in determining displacements in trusses. In **hybrid frames**, the truss members will store strain energy due to axial strains only, while the beam or frame members will store strain energy due to flexural strains only. Therefore, both **axial strains** and **flexural strains** will be considered in determining displacements in these structures.

4.1.4 Approach

The method shall be introduced using beams and the same technique shall be applied to determine frame displacements. Truss displacements shall then be discussed, and finally displacements in hybrid frames shall be considered.

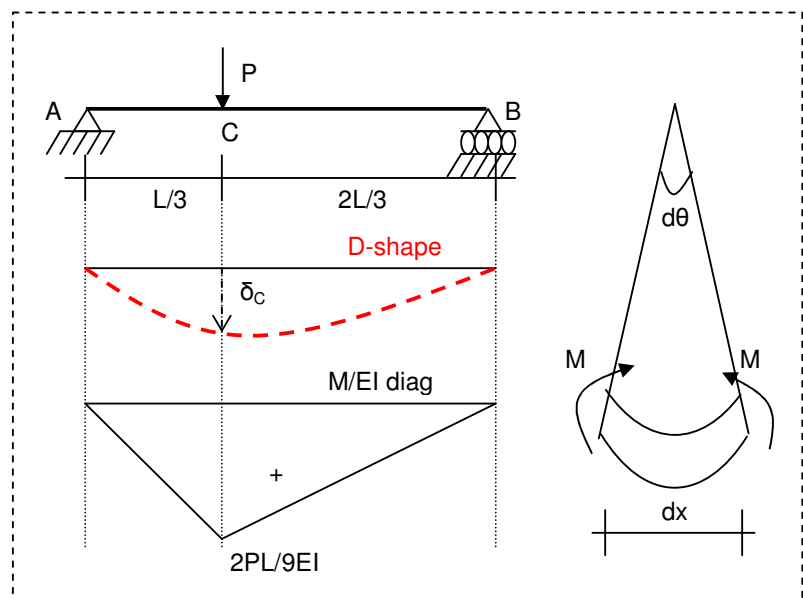
4.2 BEAM DISPLACEMENTS

4.2.1 The Particular case

Suppose it is required to determine the deflection of the point C in the beam shown below.

In an elemental section of the beam $d\theta/dx = M/EI$
so that $d\theta = (M/EI) dx$

Assuming the load P is applied gradually, the load-deflection curve and the moment-rotation curve for the moment induced in an elemental section of the beam are as shown below.



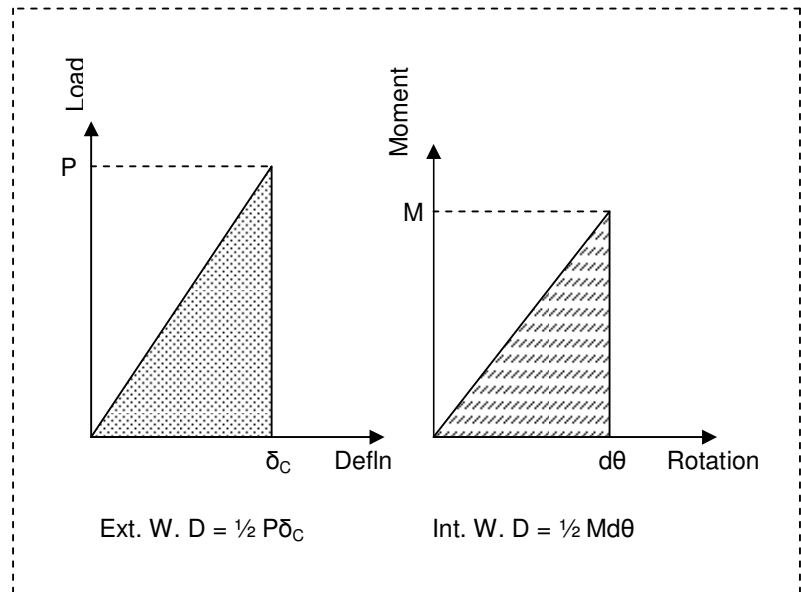
By the principle of conservation of energy: -

$$\begin{aligned}\frac{1}{2} P \delta_C &= \int_L \frac{1}{2} M d\theta \\ &= \int_L \frac{1}{2} M (M/EI) dx\end{aligned}$$

Hence

$$\frac{1}{2} P \delta_C = \int_L \frac{M^2}{2EI} dx \text{ -----Eqn 1}$$

Eqn 1 above may be used to evaluate δ_C .



NOTE: In Eqn 1, the required deflection δ_C is the deflection of the point C, which is the point of application of the load P, which is the only applied load. Eqn 1 is called the **particular solution**.

4.2.2 The General Case

Suppose it was required to determine the deflection of point D, on the beam shown below.

The solution may be achieved in four steps:

Step 1: Imagine the applied load was absent.

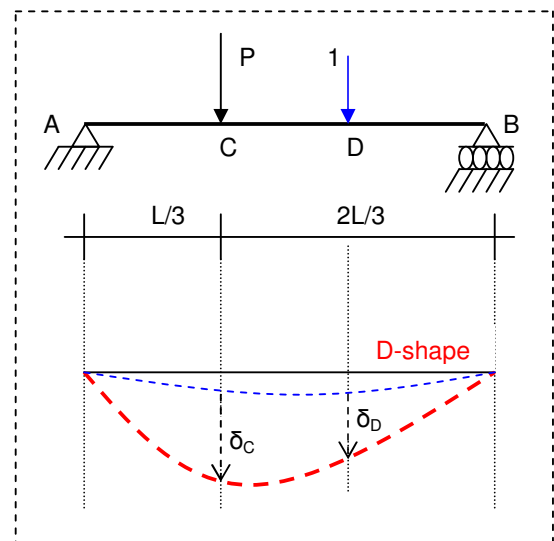
Apply a virtual unit load at the point D, in the direction and sense of interest.

The virtual unit load would induce virtual displacements which are unreal.

Step 2: With the virtual unit load in place,

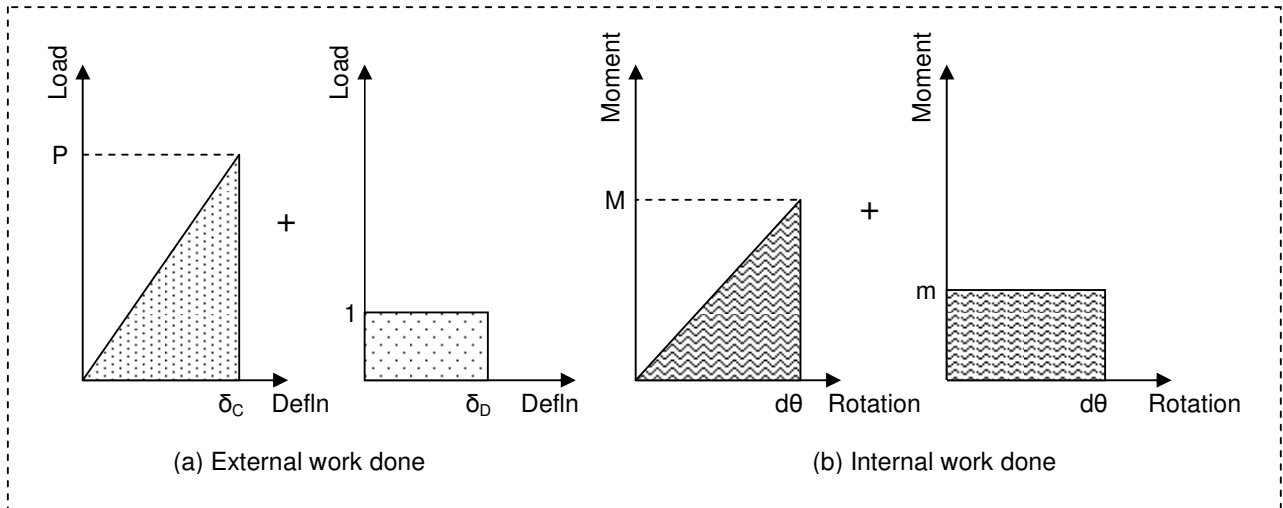
introduce the applied loading P, gradually from magnitude zero to its full magnitude.

As P rides through δ_C , the virtual unit load will ride through δ_D .



Let **m** be the bending moment at any cross-section due to the virtual unit load, and **M** be the bending moment at any cross-section due to the applied load P

Step 3: The load-deflection and moment-rotation curves for both the virtual unit load and the applied load are shown below.



Step 4: By the principle of conservation of energy: -

$$\frac{1}{2} P \delta_C + 1 \delta_D = \int_L \frac{M^2}{2EI} dx + \int_L m d\theta$$

But the particular solution requires that $\frac{1}{2} P \delta_C = \int_L \frac{M^2}{2EI} dx$

so that $\delta_D = \int_L m d\theta$, and $d\theta = (M/EI) dx$, hence

$$\delta_D = \int_L m (M/EI) dx \quad \text{-----} \quad \text{Eqn 2}$$

In general eqn (2) is written as:

$$\delta_i = \int_L m_i (M/EI) dx \quad \text{-----} \quad \text{Eqn 2*}$$

NOTE

1. In Eqn 2* m_i = a mathematical function of x for the **moment** at any cross-section due to a **virtual unit load** corresponding to the **i^{th} displacement**.
 M/EI = a mathematical function of x for the **curvature** at any cross-section due to the applied loads.
2. Eqn 2* may be executed using first principles (**ALT 1**) or using second principles (**ALT 2**). The second principles are based on **standard values of $\int m (M/EI) dx$** given in **Handout # 2**.

4.2.3 Solution Procedure

The procedure for the determination of displacements in beams using the virtual unit load method may be outlined as follows: -

1. Analyse the beam subject to the applied loads to obtain the **[M/EI] diag.**
2. For any required **ith displacement, δ_i** : -
 - * Imagine the applied loads were absent.
 - * Apply a **virtual unit load corresponding** to δ_i .
[Apply a VUL at the point of interest, in the direction and sense of interest].
 - * Analyse the beam subject to the virtual unit load only, and draw the **[m_i] diag.**
 - * Use Eqn 2* to evaluate the displacement.
3. Next required displacement.
4. Stop: END.

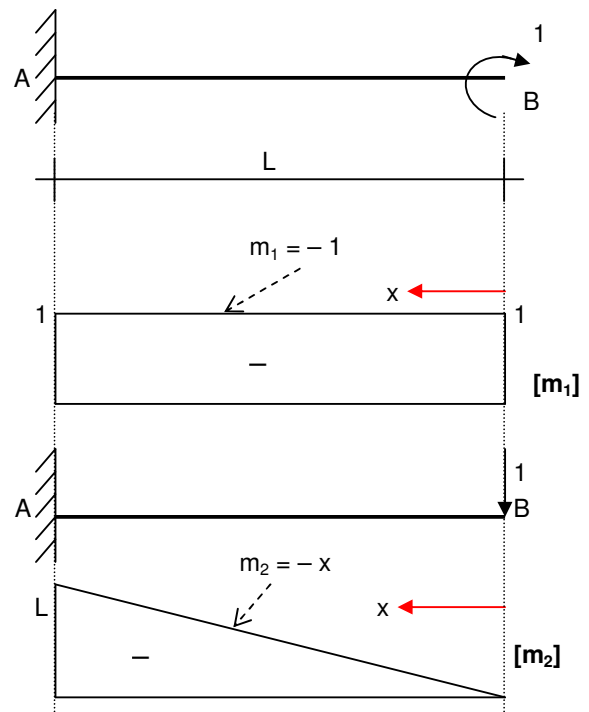
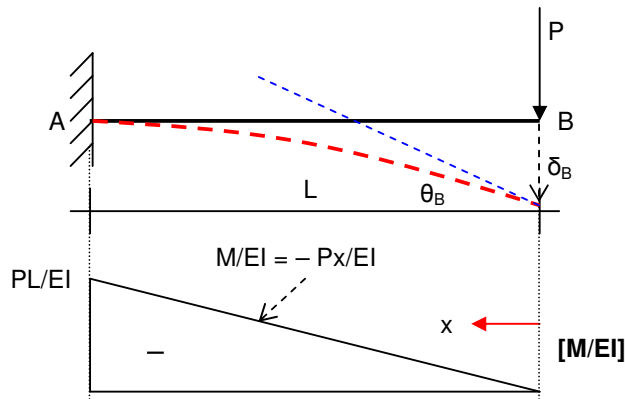
Example 1

Determine θ_B and δ_B for the cantilever shown.

Assume EI = constant

Solution

Let $\theta_B = \delta_1$ and $\delta_B = \delta_2$.



ALT 1

$$\begin{aligned}\theta_B = \delta_1 &= \int_L m_1 (M/EI) dx \\ &= \int_0^L (-1) (-Px/EI) dx = \mathbf{PL^2/2EI} \\ \delta_B = \delta_2 &= \int_L m_2 (M/EI) dx \\ &= \int_0^L (-x) (-Px/EI) dx = \mathbf{PL^3/3EI}\end{aligned}$$

ALT 2

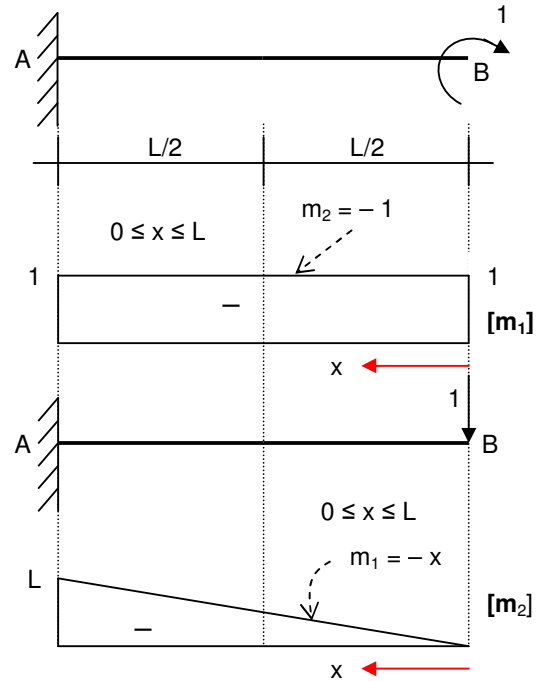
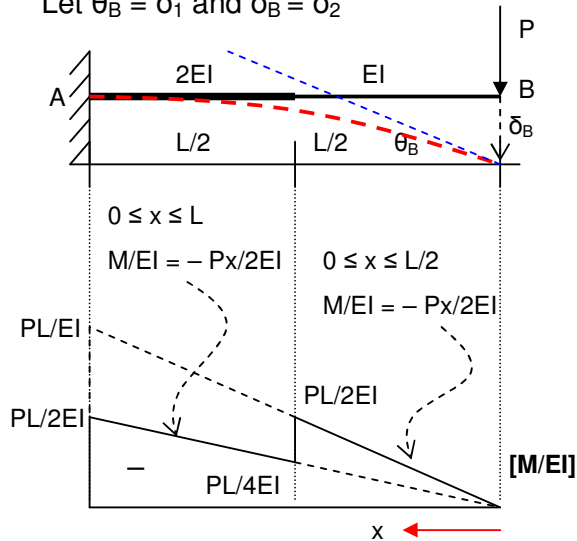
$$\begin{aligned}\theta_B &= \int_L m_1 (M/EI) dx = (1/2)acL \\ &= (1/2)(-1)(-PL/EI)(L) = \mathbf{PL^2/2EI} \\ \delta_B &= \int_L m_2 (M/EI) dx = (1/3)acL \\ &= (1/3)(-L)(-PL/EI)(L) = \mathbf{PL^3/3EI}\end{aligned}$$

Example 2

Determine θ_B and δ_B for the cantilever shown.

Solution

Let $\theta_B = \delta_1$ and $\delta_B = \delta_2$



ALT 1

$$\theta_B = \delta_1 = \int_L m_1 (M/EI) dx = \int_0^L (-1)(-Px/2EI) dx + \int_0^{L/2} (-1)(-Px/2EI) dx$$

$$= 5PL^2/16EI$$

$$\delta_B = \delta_2 = \int_L m_2 (M/EI) dx = \int_0^L (-x)(-Px/2EI) dx + \int_0^{L/2} (-x)(-Px/2EI) dx$$

$$= 3PL^3/16EI$$

ALT 2

$$\theta_B = \delta_1 = \int_L m_1 (M/EI) dx = (1/2)acL + (1/2)acL$$

$$= (1/2)(-1)(-PL/2EI)(L) + (1/2)(-1)(-PL/4EI)(L/2)$$

$$= 5PL^2/16EI$$

$$\delta_B = \delta_2 = \int_L m_2 (M/EI) dx = (1/3)acL + (1/3)acL$$

$$= (1/3)(-L)(-PL/2EI)(L) + (1/3)(-L/2)(-PL/4EI)(L/2)$$

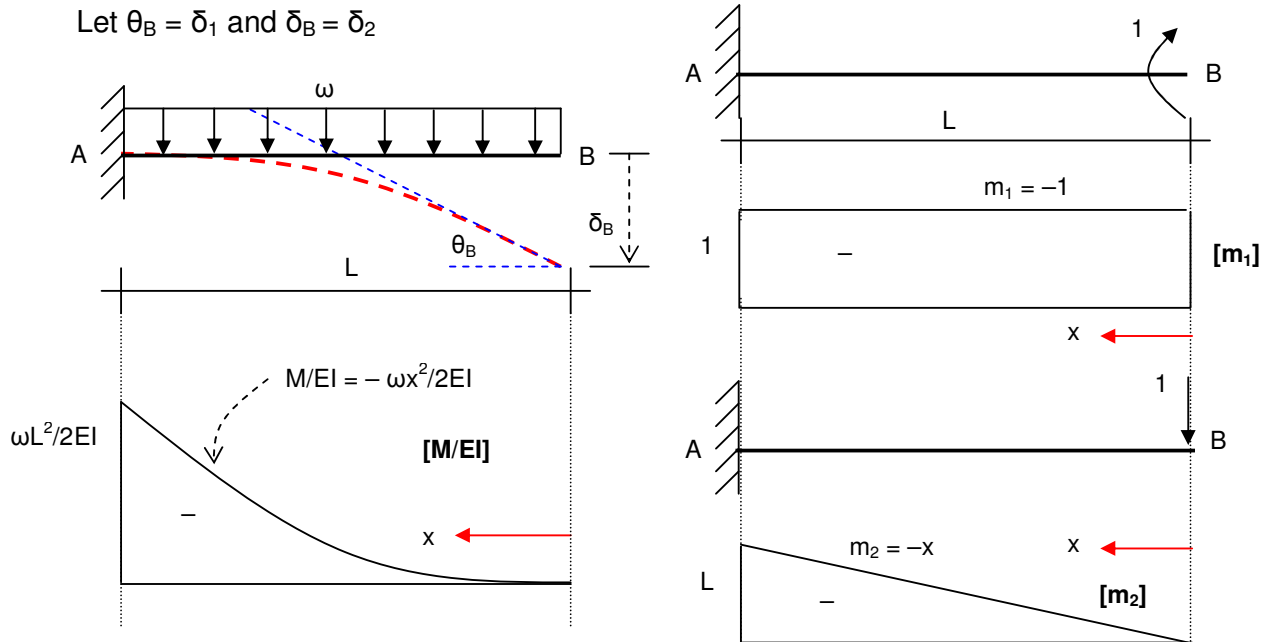
$$= 3PL^3/16EI$$

Example 3

Determine θ_B and δ_B for the beam shown. Assume $EI = \text{constant}$.

Solution

Let $\theta_B = \delta_1$ and $\delta_B = \delta_2$



ALT 1

$$\theta_B = \delta_1 = \int_L m_1 (M/EI) dx = \int_0^L (-1)(-\omega x^2/2EI) dx$$

$$= \omega L^3/6EI$$

$$\delta_B = \delta_2 = \int_L m_2 (M/EI) dx = \int_0^L (-x)(-\omega x^2/2EI) dx$$

$$= \omega L^4/8EI$$

ALT 2

$$\theta_B = \delta_1 = \int_L m_1 (M/EI) dx = (1/3)acL = (1/3)(-1)(-\omega L^2/2EI)(L)$$

$$= \omega L^3/6EI$$

$$\delta_B = \delta_2 = \int_L m_2 (M/EI) dx = (1/4)acL = (1/4)(-L)(-\omega L^2/2EI)(L)$$

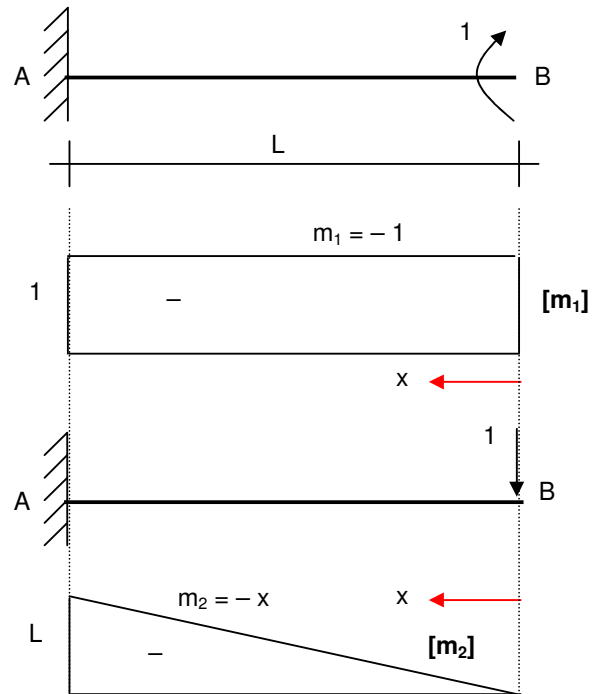
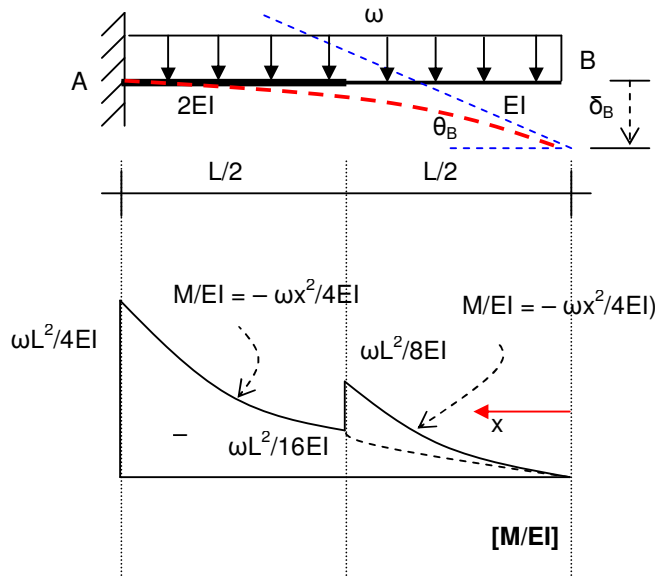
$$= \omega L^4/8EI$$

Example 4

Determine θ_B and δ_B for the beam shown.

Solution

Let $\theta_B = \delta_1$ and $\delta_B = \delta_2$



ALT 1

$$\theta_B = \delta_1 = \int_L m_1 (M/EI) dx = \int_0^L (-1)(-\omega x^2/4EI) dx + \int_0^{L/2} (-1)(-\omega x^2/4EI) dx$$

$$= 3\omega L^3/32EI$$

$$\delta_B = \delta_2 = \int_L m_2 (M/EI) dx = \int_0^L (-x)(-\omega x^2/4EI) dx + \int_0^{L/2} (-x)(-\omega x^2/4EI) dx$$

$$= 17\omega L^4/256EI$$

ALT 2

$$\theta_B = \delta_1 = \int_L m_1 (M/EI) dx = (1/3)acL + (1/3)acL$$

$$= (1/3)(-1)(-\omega L^2/4EI)(L) + (1/3)(-1)(-\omega L^2/16EI)(L/2)$$

$$= 3\omega L^3/32EI$$

$$\delta_B = \delta_2 = \int_L m_2 (M/EI) dx = (1/4)acL + (1/4)acL$$

$$= (1/4)(-L)(-\omega L^2/4EI)(L) + (1/4)(-L/2)(-\omega L^2/16EI)(L/2)$$

$$= 17\omega L^4/256EI$$

Example 5

Determine θ_A and δ_C for the beam shown alongside.

Assume $EI = \text{constant}$.

Solution

Let $\theta_A = \delta_1$ and $\delta_C = \delta_2$.

ALT 1

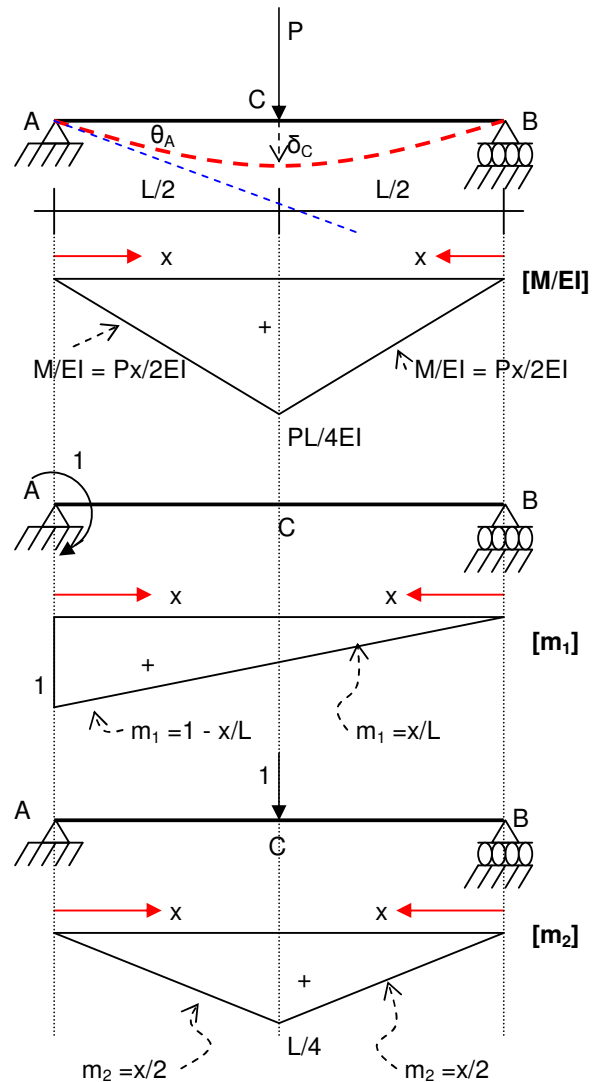
$$\begin{aligned}\theta_A = \delta_1 &= \int_L m_1 (M/EI) dx \\ &= \int_0^{L/2} (1 - x/L) (Px/2EI) dx \\ &\quad + \int_0^{L/2} (x/L) (Px/2EI) dx \\ &= PL^2/16EI\end{aligned}$$

$$\begin{aligned}\delta_C = \delta_2 &= \int_L m_2 (M/EI) dx \\ &= 2 \int_0^{L/2} (x/2) (Px/2EI) dx \\ &= PL^3/48EI\end{aligned}$$

ALT 2

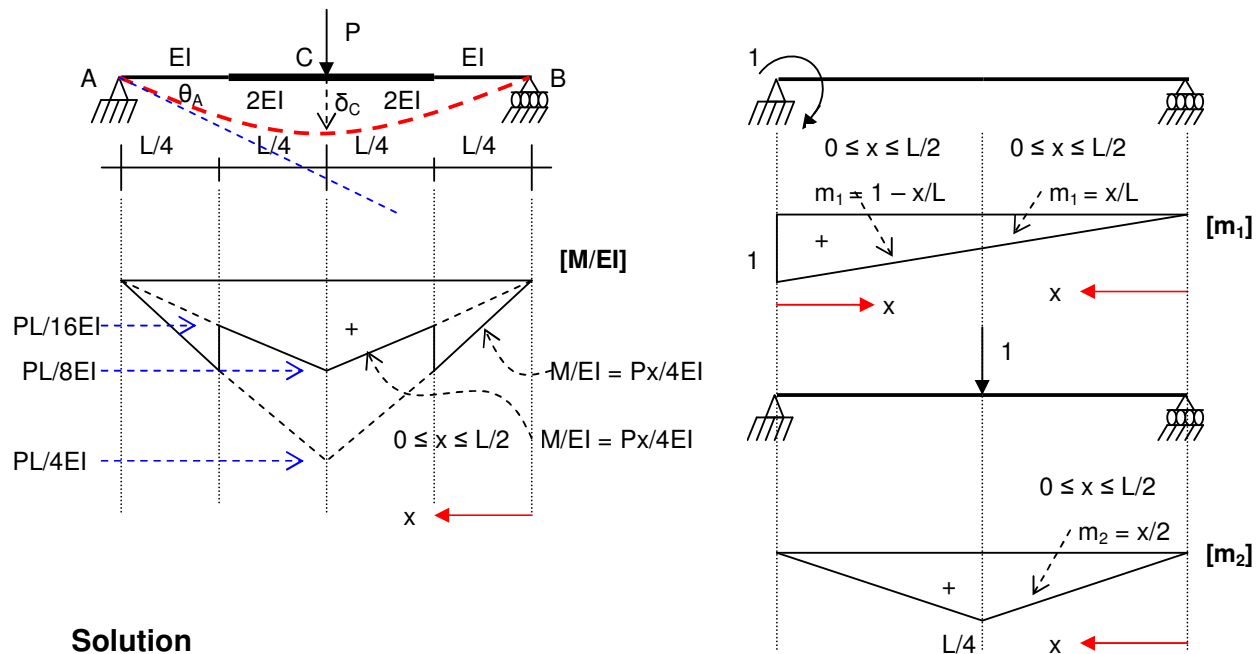
$$\begin{aligned}\theta_A = \delta_1 &= \int_L m_1 (M/EI) dx \\ &= (1/3)acL + (1/6)cL(2a + b) \\ &= (1/3)(1/2)(PL/4EI)(L/2) \\ &\quad + (1/6)(PL/4EI)(L/2)(2(1/2) + 1) \\ &= PL^2/16EI\end{aligned}$$

$$\begin{aligned}\delta_C = \delta_2 &= \int_L m_2 (M/EI) dx \\ &= (2)(1/3)acL \\ &= (2)(1/3)(L/4)(PL/4EI)(L/2) \\ &= PL^3/48EI\end{aligned}$$



Example 6

Determine θ_A and δ_C for the beam shown below.



Solution

Let $\theta_A = \delta_1$ and $\delta_C = \delta_2$

ALT 1

$$\begin{aligned}\theta_A = \delta_1 &= \int_L m_1 (M/EI) dx = \int_0^{L/2} (1 - x/L)(Px/4EI) dx + \int_0^{L/2} (x/L)(Px/4EI) dx \\ &\quad + \int_0^{L/4} (1 - x/L)(Px/4EI) dx + \int_0^{L/4} (x/L)(Px/4EI) dx \\ &= 5PL^2/128EI\end{aligned}$$

$$\begin{aligned}\delta_C = \delta_2 &= \int_L m_2 (M/EI) dx = 2\left\{\int_0^{L/2} (x/2)(Px/4EI) dx + \int_0^{L/4} (x/2)(Px/4EI) dx\right\} \\ &= 3PL^3/256EI\end{aligned}$$

ALT 2

$$\begin{aligned}\theta_A = \delta_1 &= \int_L m_1 (M/EI) dx = (1/6)cL(2a+b) + (1/3)acL + (1/6)cL(2a+b) + (1/3)acL \\ &= (1/6)(PL/8EI)(L/2)(1+1) + (1/3)(1/2)(PL/8EI)(L/2) \\ &\quad + (1/6)(PL/16EI)(L/4)(6/4+1) + (1/3)(1/4)(PL/16EI)(L/4) \\ &= 5PL^2/128EI\end{aligned}$$

$$\begin{aligned}\delta_C = \delta_2 &= \int_L m_2 (M/EI) dx = 2(1/3)acL + 2(1/3)acL \\ &= 2(1/3)(L/4)(PL/8EI)(L/2) + 2(1/3)(L/8)(PL/16EI)(L/4) \\ &= 3PL^3/256EI\end{aligned}$$

Example 7

Determine θ_A and δ_C for the beam shown.

Assume $EI = \text{constant}$

Solution

Let $\theta_A = \delta_1$ and $\delta_C = \delta_2$

ALT 1

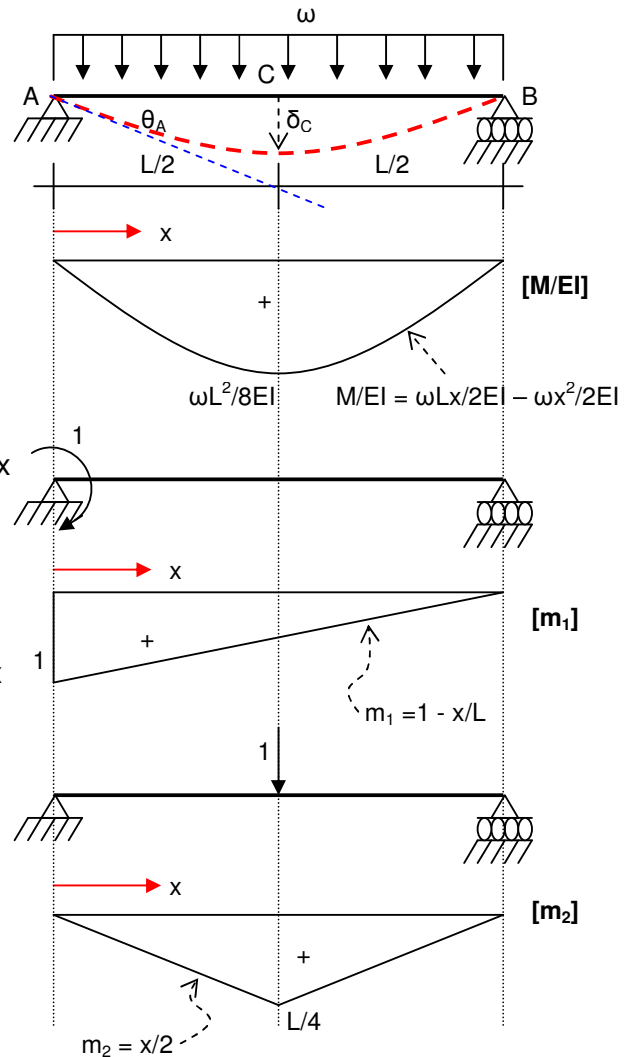
$$\begin{aligned}\theta_A = \delta_1 &= \int_L m_1 (M/EI) dx \\ &= \int_0^L (1 - x/L) (\omega Lx/2EI - \omega x^2/2EI) dx \\ &= \omega L^3/24EI\end{aligned}$$

$$\begin{aligned}\delta_C = \delta_2 &= \int_L m_2 (M/EI) dx \\ &= 2 \int_0^{L/2} (x/2) (\omega Lx/2EI - \omega x^2/2EI) dx \\ &= 5\omega L^4/384EI\end{aligned}$$

ALT 2

$$\begin{aligned}\theta_A &= \int_L m_1 (M/EI) dx \\ &= (1/12)cL(5a+3b) + (5/12)acL \\ &= (1/12)(\omega L^2/8EI)(L/2)(5(1/2)+3(1)) \\ &\quad + (5/12)(1/2)(\omega L^2/8EI)(L/2) \\ &= \omega L^3/24EI\end{aligned}$$

$$\begin{aligned}\delta_B &= \int_L m_2 (M/EI) dx = 2(5/12)acL \\ &= 2(5/12)(L/4)(\omega L^2/8EI)(L/2) \\ &= 5\omega L^4/384EI\end{aligned}$$



Example 8

Determine θ_A and δ_C for the beam shown below.

Solution

Let $\theta_A = \delta_1$ and $\delta_C = \delta_2$

ALT 1

$$\begin{aligned}\theta_A = \delta_1 &= \int_L m_1 (M/EI) dx \\ &= \int_0^L (1 - x/L) (\omega Lx/4EI - \omega x^2/4EI) dx \\ &\quad + \int_0^{L/4} (1 - x/L) (\omega Lx/4EI - \omega x^2/4EI) dx \\ &\quad + \int_0^{L/4} (x/L) (\omega Lx/4EI - \omega x^2/4EI) dx \\ &= 7\omega L^3/256EI\end{aligned}$$

$$\begin{aligned}\delta_C = \delta_2 &= \int_L m_2 (M/EI) dx \\ &= 2 \left\{ \int_0^{L/2} (x/2) (\omega Lx/4EI - \omega x^2/4EI) dx \right. \\ &\quad \left. + \int_0^{L/4} (x/2) (\omega Lx/4EI - \omega x^2/4EI) dx \right\} \\ &= 31\omega L^4/4096EI\end{aligned}$$

ALT 2

Upper M/EI is NOT standard,
ALT 2 CANNOT work.

